

Frostman data on $E \subset \mathbb{R}^d$, with $s > S = d/2$

AUXILIARY SPACES: DIRECTIONS

$\mathbb{R}^{[S]+1}$: radial L^p estimate
 Starting cap exponent $[S]$
 Odd $d=2k-1$: upgrade in $\mathbb{R}^{k+1}$
 $k-1 \rightarrow S$

Lift cap bounds;
 pull back pair filters

ORIGINAL $\mathbb{R}^d$: MEASURES

Positive separated restrictions
 Dyadic classes and profiles
 $A_{j,n} = m_j(n/N) - Sn/N$

Original-space tests and smooth angular masks
 S -cap bounds after power-small pair deletion; $\Gamma_N \downarrow \Gamma_\infty$

DISTANCE ANALYSIS

Pairs remain in
 $E \times E$

Distance shells: $D(x, y) = |x - y|$ in $\mathbb{R}^d$
 Stationary phase and the graph estimate

State recursion + two-depth potential + entry
 Restore original masses $\Rightarrow$ summable shell decay

SCALAR MONOTONE-SHELL LIMIT
 $\alpha_N \downarrow \alpha_\infty \neq 0$ and shell approximants τ_N
 $\alpha_\infty \ll \mathcal{L}^1$, supported on $\Delta(E) \Rightarrow \mathcal{L}^1(\Delta(E)) > 0$